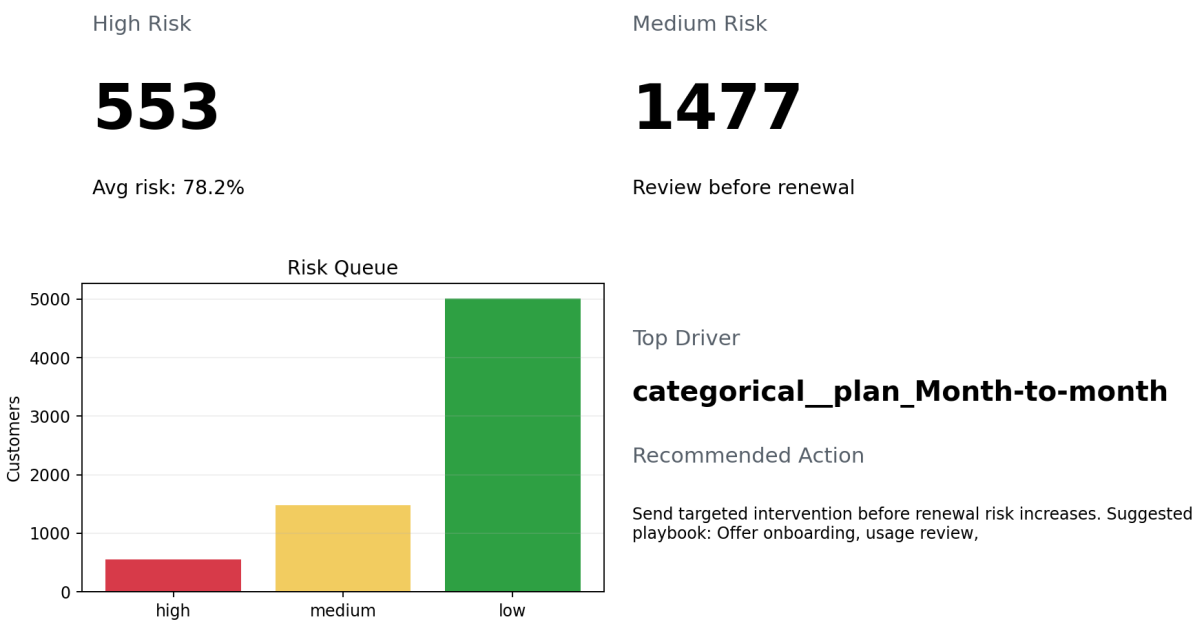


Churn Prediction + Retention Report

Executive summary: this report ranks customers by churn risk and turns model output into a prioritized retention action queue.

The model ranks customers by churn risk. The top-risk segment should be reviewed first because it concentrates the highest expected churn probability.

Churn Retention Executive Snapshot



Dataset Summary

Metric	Value
rows	7043
columns	11
observed_churn_rate	26.54%
duplicate_customer_ids	0
leakage_warnings	0

Model Metrics

Metric	Value
roc_auc	0.8435
pr_auc	0.6564
baseline_pr_auc	0.2655
accuracy	0.7908

Metric	Value
precision	0.6319
recall	0.508
f1	0.5632
confusion_matrix	[[1386, 166], [276, 285]]

Risk Segment Breakdown

risk_segment	customers	avg_churn_probability
high	553	0.7818
medium	1477	0.5405
low	5013	0.1267

Lift Table Interpretation

The highest-risk 10% captures 30.1% of observed churners. The highest-risk 20% captures 51.5%. Review the 553 high-risk customers first.

decile	customers	churners	churn_rate	avg_churn_probability	cumulative_churners	cumulative_churn_capture_rate
1	705	562	0.7972	0.7599	562	0.3007
2	704	401	0.5696	0.5845	963	0.5152
3	704	324	0.4602	0.4489	1287	0.6886
4	704	231	0.3281	0.3236	1518	0.8122
5	705	156	0.2213	0.2268	1674	0.8957

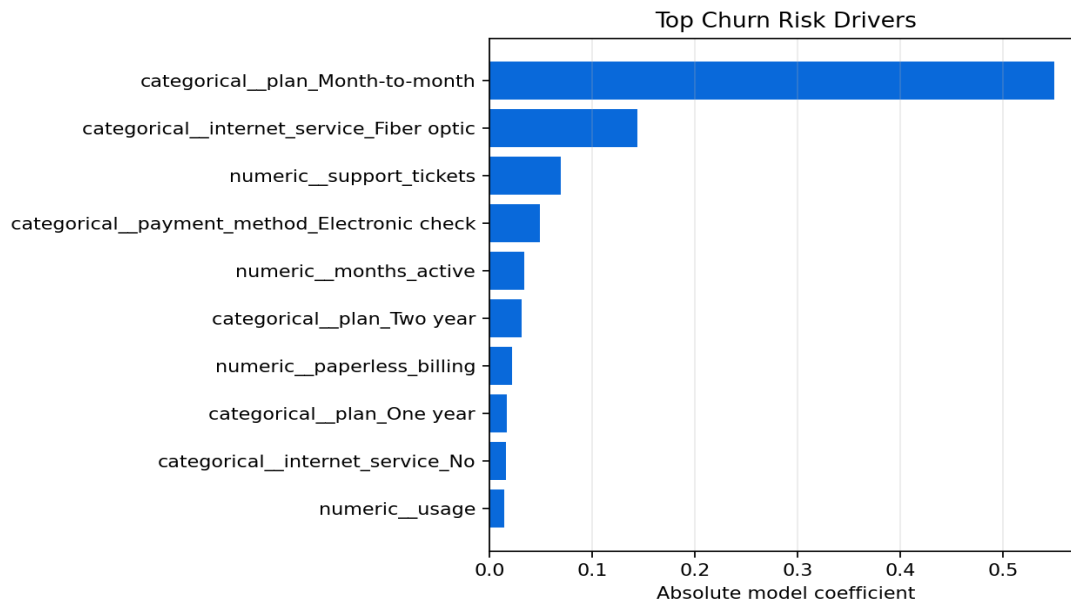
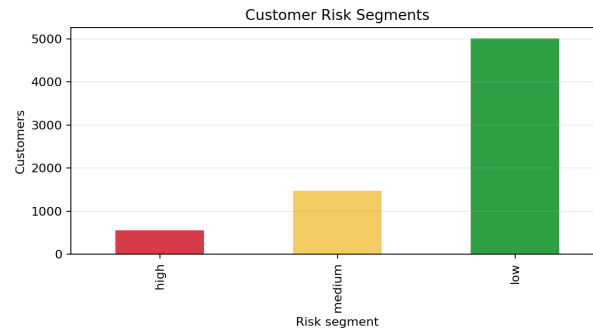
Top Recommended Actions

customer_id	churn_probability	likely_drivers	recommended_action
1	0.6824	usage, support_tickets, payments	Send targeted intervention before renewal risk increases. Suggested playbook: Offer onboarding, usage review, or workflow training to new users.
2	0.0442	usage	Monitor monthly and reinforce the strongest positive behavior: usage.
3	0.5096	usage, support_tickets, payments	Send targeted intervention before renewal risk increases. Suggested playbook: Offer onboarding, usage review, or workflow training to new users.
4	0.0398	usage	Monitor monthly and reinforce the strongest positive behavior: usage.
5	0.6535	support_tickets, payments	Send targeted intervention before renewal risk increases. Suggested playbook: Prioritize proactive support outreach and assign a customer success manager.

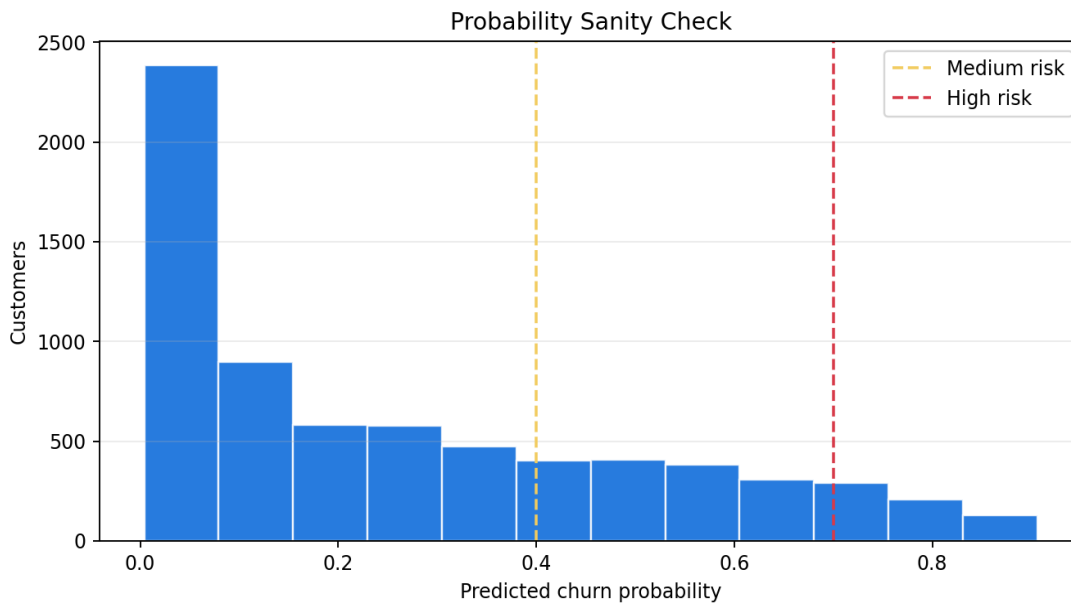
Recommended Retention Strategy

Use the retention recommendation CSV as the operating queue. High-risk customers should receive direct outreach; medium-risk customers should receive targeted education, usage review, or plan-fit checks.

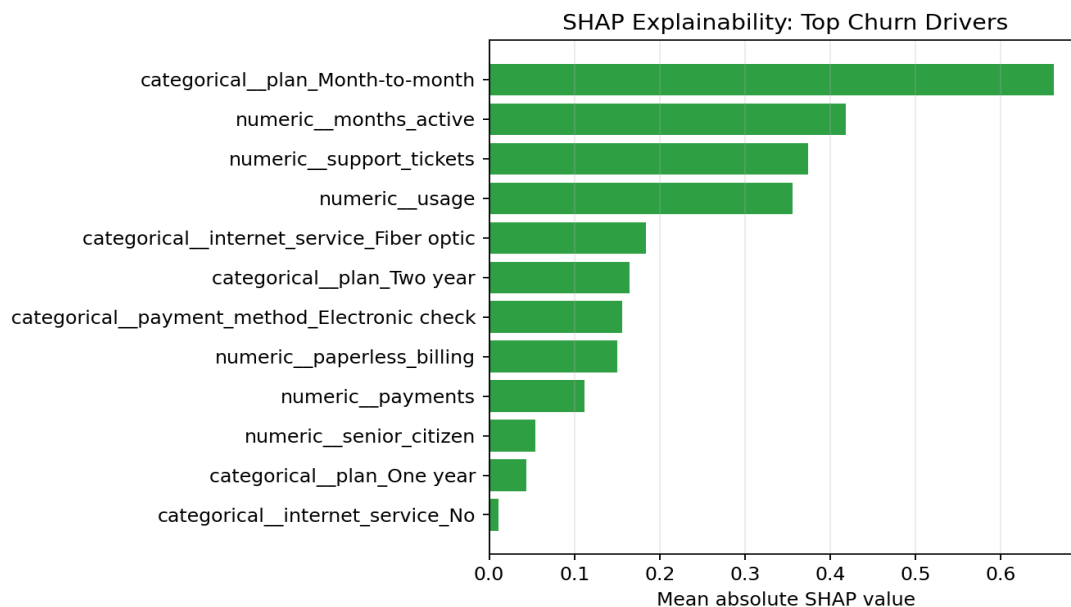
Visual Evidence



Probability Sanity Check



SHAP Explainability



Limitations

This report does not guarantee a specific accuracy number. Results depend on data volume, feature quality, label quality, and whether post-churn leakage columns were excluded before modeling.

Next Steps

Recommended next steps: review leakage warnings, confirm the business definition of churn, enrich weak features, run the action queue, and compare retention outcomes after the next billing cycle.

Scope and Privacy

This package is for a one-time churn analysis and retention report from a provided CSV. API deployment, recurring automation, CRM integration, and dashboard hosting are custom add-ons. Client data should not be committed to a public repository.